

RODRIGO GOMES DE ARAÚJO

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SUMMARY

MSc Computer Science student (graduating **July 2027**) focused on systems where **correctness and robustness** are key, such as protocols, consensus and infrastructure people actively rely on. Built a **proof-of-work blockchain** in Rust and is currently extending session types to **enforce correct usage of cryptographic resources** at compile-time, with thesis completion expected in July 2027. Works well **autonomously** and **asynchronously** and emphasizes **testing, documentation and API** in the development process. Comfortable working remotely and async. Based in Porto, Portugal.

EDUCATION

Master's <i>Computer Science</i>	Sep 2025 – Present (Jul 2027) <i>University of Porto (FCUP)</i>
Bachelor's <i>Computer Science and Engineering</i>	Sep 2022 – Jul 2025 <i>University of Porto (FEUP)</i>

RESEARCH

CaST (Cryptographically-aware Session Types) <i>Master's Thesis</i>	Jul 2026 – Present (Jul 2027) <i>FreeST, Haskell, Type Systems, Applied Cryptography</i>
- Extending the expressiveness of session types to capture cryptographic resource usage, so that its misuse in protocols is detected and rejected at compile-time. Resource tracking composes across nested subprotocols of arbitrary depth and is preserved under sequential composition and recursion. - Formalizing the extension and providing a soundness proof, integrating it into the FreeST compiler and applying it to smart contracts and protocol specification.	

PROJECTS

PoW Blockchain for Decentralized Auctions | Rust, Tokio, LibP2P, Dalek, Docker

- Extensive **research on the state of the art of Bitcoin**, Ethereum and Solana and investigation regarding the security and resilience to failure of these systems.
- Developed a DHT overlay based on Kademlia for **decentralized network construction**.
- PoW Blockchain with **Blake2b hashing**, **Ed25519 signatures** and **Merkle Trees** for transaction validation.
- **Client for auctions** using the blockchain layer as a public ledger.
- Docker **adversarial network simulation** with clients and bootstrap, well behaved and malicious nodes.
- Developed as Data Systems Security final project and graded 19.1/20.

Real-Time Modular Synthesizer | Rust, Tokio, Rodio, Ratatui, SQLite

- **Real-time audio engine** isolated from the UI and control logic, communicating through **lock-free atomics and message passing**. The audio path never locks, allocates or blocks on I/O.
- **Modular**, leveraging Rust's type system. The audio DSP graph is built at runtime and instantiated per note, so voices run **independently** and the sound can be **reconfigured in real-time** without interruption.

AWARDS

2nd Place in IEEE RetroJam 2025 IEEE UP Student Branch	Oct 2025
Built a game from scratch in 48 hours, using Rust and Raylib, learning how to reduce a problem to its core requirements.	
Academic Merit Awards	2020, 2021, 2022
Participation in RoboCup 2016 RoboCup Federation	Jun 2016
1st Place in the National Robotics Championship Instituto Politécnico de Bragança	May 2016

TECHNICAL SKILLS

Programming Languages: Rust, Haskell, C/C++, OCaml, Java, SQL
Frameworks & Tools: Git, UNIX, Tokio, Docker, Plotly, POSIX, STM, gRPC, LibP2P, Dalek, PostgreSQL, Terraform, Raylib and OpenMP
Spoken Languages: Portuguese (Native), English (B2 First, Cambridge), Spanish (Proficient)